

## Full Length Article

# DPGR-Net: A disentangled population-guided graph neural network for major depressive disorder identification

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## ABSTRACT

Major depressive disorder (MDD) is a brain disorder characterized by substantial individual heterogeneity, and its accurate diagnosis remains challenging in clinical practice. Graph neural networks applied to functional brain networks (FBNs) constructed from resting-state functional magnetic resonance imaging (rs-fMRI) data have shown promise for MDD diagnosis. However, in multicenter settings, site differences are often entangled with disease-related information, which not only interferes with the model's learning of disease patterns but also compromises the reliable modeling of topological patterns in individual FBNs, thereby limiting identification performance. To address this problem, we propose a disentangled population-guided graph neural network with dynamic refinement (DPGR-Net) for MDD identification. Specifically, DPGR-Net first constructs individual FBN from rs-fMRI data and learns discriminative subject-level brain graph representations. To mitigate the interference of site effects, we disentangle disease-related associations from site-related variations at the population level and impose an independence constraint to learn more stable cross-site disease-discriminative representations. On this basis, we innovatively design a population-guided dynamic optimization mechanism for individual brain graphs, which feeds the stable disease-discriminative information learned at the population level back into the structural updating of individual brain graphs, thereby enhancing disease-related connections while suppressing site-related noisy connections. Experimental results on the public REST-meta-MDD dataset show that the proposed method achieves an ACC of 92.37% and outperforms existing state-of-the-art methods. Furthermore, DPGR-Net identifies discriminative brain regions and aberrant functional connectivity patterns associated with MDD, thereby advancing the interpretability and clinical utility of rs-fMRI for MDD diagnosis.

## 1. Introduction

Major depressive disorder (MDD) is a disabling mental disorder that severely impairs patients' cognitive functioning and social adaptability (Mulders et al., 2015). Studies have shown that the neuropathological mechanisms of MDD are closely associated with complex abnormalities in brain function and exhibit marked individual heterogeneity (Noman et al., 2024). However, current clinical diagnosis mainly relies on symptom-based assessment and clinicians' judgment, lacking an objective characterization of the underlying neurobiological abnormalities (Zhang et al., 2023). This limits the ability to capture complex disease-related functional changes and individual differences.

In recent years, resting-state functional magnetic resonance imaging (rs-fMRI) brain network analysis has become an important approach for studying neuropathological mechanisms and assisting clinical diagnosis (Ganaie & Tanveer, 2024; Malik et al., 2024; Tanveer et al., 2023).

In such studies, regions of interest (ROIs) are typically defined as network nodes, and functional connectivity (FC) networks are constructed based on statistical dependencies among blood-oxygen-level-dependent (BOLD) signals between regions, thereby characterizing individual-level patterns of brain functional organization (Li et al., 2020; Liu et al., 2024; Yan et al., 2025). With the rapid development of graph neural networks (GNNs) for modeling non-Euclidean structured data (Li et al., 2022; Si et al., 2025; Tanveer et al., 2022), individual brain graph-based analysis has become an important paradigm for brain disorder identification. For example, BrainGNN (Li et al., 2021) introduces ROI-aware graph convolution and node selection mechanisms on individual brain graphs, which not only improve brain disorder identification performance but also enhance the interpretability of key brain regions and their connectivity patterns. Furthermore, Zhang et al. (2026) proposed an individual brain graph analysis method based on graph contrastive learning. By constructing a dual-view topological augmentation

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strategy, their method improves the model's ability to represent differences in FC across individuals, thereby providing a more effective representation learning framework for identifying abnormal brain network patterns in brain disorder diagnosis. These methods learn topological representations from subject-level functional brain networks (FBNs), enabling a more effective use of the topological organization of brain networks and providing effective individualized representations for brain disorder identification (Jiang et al., 2026; Ma et al., 2023). However, their high performance often relies on relatively small-scale single-center datasets acquired under consistent conditions (Ren et al., 2025). In clinical practice, large-scale rs-fMRI data are often collected from multiple centers. Due to differences in scanners, acquisition protocols, and preprocessing procedures, multi-center rs-fMRI data often exhibit site effects, which may become entangled with disease-related patterns (Lin et al., 2023). Therefore, models trained on single-center data are vulnerable to site heterogeneity, leading to reduced generalizability and unreliable biological interpretations.

To better utilize multi-center data, population graph frameworks have been proposed (Cao et al., 2021). In such frameworks, subjects are represented as nodes with features derived from individual imaging representations. Edges are constructed from non-imaging information such as age, gender and site to model inter-subject similarity (Hao et al., 2025). By using non-imaging information as complementary relational priors, these methods reduce reliance on heterogeneous imaging features, thereby improving feature discriminability and better capturing disease-related patterns shared across sites. Parisot et al. (2017) proposed PopulationGCN, the earliest method for population-graph brain disorder prediction. It integrates imaging features and phenotypic information into a unified graph, laying the foundation for population-based brain analysis. Huang and Chung (2020) proposed EV-GCN, which improved the flexibility and robustness of population relationship modeling through adaptive graph learning and variational edge modeling. Furthermore, Cao et al. (2021) proposed a DeepGCN-based multi-site population graph framework to capture higher-order inter-subject associations, thus enhancing the representational capacity of population graphs. By modeling inter-subject relationships, these methods improve disease identification performance. However, they mainly learn shared subject-level patterns, while their node representations are typically compressed into subject-level feature vectors, making it difficult to capture the fine-grained differences embedded in individual brain connectivity patterns (Gu et al., 2025). Such differences are critical for MDD identification due to its substantial individual heterogeneity (Kogonkar et al., 2014).

Therefore, some studies have further integrated individual and population graphs to preserve topological features of individual brain networks while fully exploiting cross-subject relational information, thereby improving brain disorder identification accuracy. For example, Zhang et al. (2023) proposed LG-GNN, a local-to-global joint modeling framework that identifies key regional biomarkers from individual brain graphs and further captures inter-subject relationships on a population graph using non-imaging information. Furthermore, Gu et al. (2025) proposed FC-HGNN, which designs a convolutional mechanism to integrate intra- and inter-hemispheric information flow, to characterize the coordinated interaction patterns between the left and right hemispheres, while also modeling cross-subject relationships on a heterogeneous population graph. Such methods jointly leverage topological information from individual brain networks and relational information at the population level, thereby providing a new modeling perspective for brain disorder identification. Although these methods can jointly model multi-site samples, they usually lack an explicit mechanism to disentangle disease-related information from site-related variations. Site bias in multi-site data is often highly entangled with disease-related distributional differences and may even dominate true neuropathological variations (Lin et al., 2023). As a result, the learned representations may still contain site-related confounders, limiting the model's ability to capture MDD-related neurobiological characteristics (Zhao & Zhang,

2024). Since individual brain graphs are typically constructed from FC matrices (Li et al., 2021), site heterogeneity may affect both subject-level feature distributions and individual graph topologies (Song et al., 2024). If such site-related noise is not properly addressed at the individual graph level, its influence may propagate during graph learning, further weakening individual representation learning and population-level relationship modeling.

Therefore, we propose a multi-site MDD identification framework, termed Disentangled Population-Guided Refinement Graph Neural Network (DPGR-Net). DPGR-Net learns shared disease-related representations from cross-subject relationships and feeds them back to refine individual FC topology, thereby improving task-guided individual graph modeling while reducing the influence of site-related variations. Specifically, at the individual level, subject-specific FBNs are constructed from rs-fMRI data to learn initial brain graph representations. At the population graph level, a disentangled dual-channel population graph (DDPG) module is designed to separately model disease-related associations and site-related variations through a disease graph and a site graph. An independence constraint is further imposed to reduce the statistical dependence between the two channels, thereby alleviating the interference of site effects in disease representation learning. On this basis, we innovatively propose a population-guided dynamic refinement mechanism (PDRM) for individual brain networks. PDRM feeds shared disease-related patterns learned at the population level back to the individual graph level to adaptively refine FC topology. This bidirectional individual-population interaction enables task-guided topology modulation of subject-specific FBNs while reducing task-irrelevant variations. Overall, the main contributions are summarized as follows:

1. We propose DPGR-Net, for multi-site MDD identification, which explicitly separates disease-related information from site-related interference through DDPG modelling.
2. We innovatively design the PDRM module to use shared population-level disease representations to adaptively refine individual FC topology, thereby incorporating task-guided topology adjustments while reducing the influence of task-irrelevant variations.
3. Extensive experiments on the Rest-Meta-MDD dataset demonstrate that the proposed method outperforms state-of-the-art approaches and exhibits strong robustness in multi-site settings.

## 2. Method

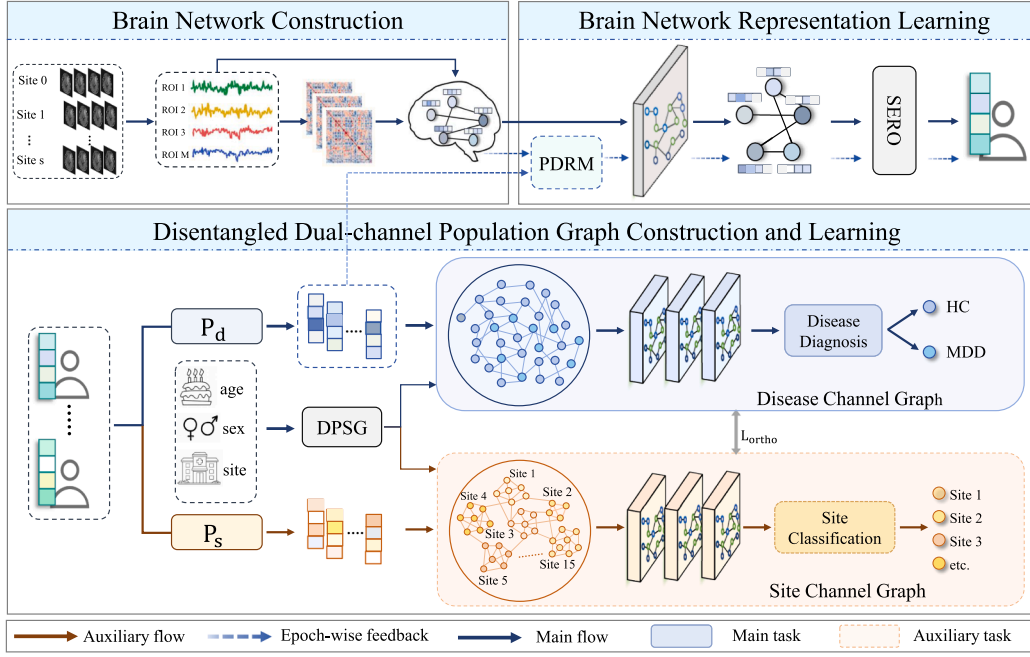
Overall, DPGR-Net follows a bidirectional individual-population modeling strategy. First, individual FBNs are constructed from rs-fMRI data to characterize individual connectivity patterns. A graph encoder is then used to learn subject-level brain network representations. To mitigate multi-site heterogeneity, DDPG is constructed to separately model disease-related information and site-related variations. Finally, population-level disease-related patterns are fed back to refine individual FC topology, enabling task-guided topology modulation while reducing the influence of task-irrelevant variations, as illustrated in Fig. 1.

### 2.1. Brain network construction

First, individual FBNs are constructed from rs-fMRI data. For each subject, the brain is parcellated into  $M$  ROIs, forming the node set  $V = \{v_1, v_2, \dots, v_M\}$ . The  $m$ th ROI is represented by a BOLD time series of length  $T$ , denoted as  $\mathbf{x}_m = [x_m^1, x_m^2, \dots, x_m^T]$ . These BOLD time series characterize the functional activity patterns of different brain regions and serve as the basis for estimating FC between ROIs. To quantify the functional similarity between any two ROIs  $u$  and  $v$ , their FC strength is calculated using the Pearson correlation coefficient:

$$r_{uv} = \frac{\sum_{t=1}^T (x_u^t - \bar{x}_u)(x_v^t - \bar{x}_v)}{\sqrt{\sum_{t=1}^T (x_u^t - \bar{x}_u)^2} \sqrt{\sum_{t=1}^T (x_v^t - \bar{x}_v)^2}} \quad (1)$$

where  $\bar{x}_u$  and  $\bar{x}_v$  are the mean BOLD signals of ROIs  $u$  and  $v$ , respectively.



**Fig. 1.** Individual FBNs are first constructed from rs-fMRI data and encoded by a GIN with SERO readout to obtain subject-level graph representations. These representations are then projected into disease- and site-related subspaces by  $P_d$  and  $P_s$ , respectively, and combined with non-imaging information in the DPSG module to generate the edge weights for DDPG construction. The disease-discriminative population information is further fed back by PDRM to dynamically refine individual brain graph structures.

Pairwise ROI correlations form an FC matrix  $\mathbf{R} \in \mathbb{R}^{M \times M}$  for each subject. Fisher's z-transformation (Yuan et al., 2013) is then applied to stabilize the variance and improve normality, yielding the transformed matrix  $\mathbf{C}$ :

$$c_{uv} = \frac{1}{2} \ln \left( \frac{1 + r_{uv}}{1 - r_{uv}} \right) \quad (2)$$

The  $m$ th row of  $\mathbf{C}$  is further used as the initial feature vector  $\mathbf{x}'_m$  of node  $v_m$ . To reduce redundant weak connections, a thresholding strategy is applied to the matrix  $\mathbf{C}$ . Only edges with correlation values greater than a predefined threshold  $\tau$  are retained to construct the initial adjacency matrix  $\mathbf{A}_{orig}$ . Accordingly, an individual-level brain functional graph  $G = (V, E)$  is constructed for each subject, where  $E$  is determined by the thresholded FC matrix.

It should be noted that this step only provides an initial structural representation, and the resulting individual brain graphs may still contain noisy, redundant, or task-irrelevant connections (Zhang et al., 2026). To address this, a PDRM module is introduced to dynamically refine the network structure, as detailed in Section 2.4.

## 2.2. Brain network representation learning

To learn discriminative FBN representations, we adopt Graph Isomorphism Network (GIN) (Xu et al., 2019) to encode individual brain graphs. Given the heterogeneous neighborhood patterns across different ROIs, GNNs based on simple average aggregation may fail to capture complex structural differences effectively. With learnable parameters and a multi-layer perceptron (MLP), GIN better distinguishes diverse neighborhood structures and learns more discriminative individual FBN representations (Thapaliya et al., 2025a). Specifically, at the  $k$ th layer, GIN updates each node representation by aggregating its own and neighboring features. The update rule is defined as:

$$\mathbf{h}_m^{(k)} = \text{MLP}^{(k)} \left( (1 + \epsilon^{(k)}) \mathbf{x}'_v^{(k-1)} + \sum_{u \in N(m)} \mathbf{x}'_u^{(k-1)} \right) \quad (3)$$

where  $\mathbf{h}_m^{(k)}$  denotes the representation of node  $m$  at the  $k$ th layer,  $N(m)$  represents the set of neighboring nodes of node  $v$ ,  $\epsilon^{(k)}$  is a learnable parameter, and  $\text{MLP}^{(k)}(\cdot)$  denotes the MLP at the  $k$ th layer.

After  $K$  layers of GIN learning, the updated node feature set can be represented as  $\mathbf{H} = \{\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_M\} \in \mathbb{R}^{M \times d}$ .

In FBNs, ROIs often contribute unequally to disease discrimination. Some disease-related regions may exhibit more pronounced functional alterations and therefore provide more informative cues for identifying neurological disorders (Gu et al., 2025). To emphasize these regions, we adopt the Squeeze-Excitation Readout (SERO) mechanism (Kim et al., 2021) to model ROI importance. Specifically, given the node representation  $\mathbf{h}_m$ , the importance score of each ROI is computed using a learnable attention function:

$$\alpha_m = \mathbf{w}^T \sigma(\mathbf{W} \mathbf{h}_m^T + \mathbf{b}) \quad (4)$$

where  $\mathbf{W} \in \mathbb{R}^{d' \times d}$  and  $\mathbf{b} \in \mathbb{R}^{d'}$  are the weight matrix and bias vector of the attention layer,  $\sigma(\cdot)$  denotes ReLU activation function, and  $\mathbf{w} \in \mathbb{R}^{d'}$  is a learnable projection vector that compresses the hidden representation into a scalar score.

To ensure comparability across the entire brain network, these scores are normalized using a Softmax function to obtain the attention weights  $\tilde{\alpha}_m$ . The node features are then aggregated in a weighted manner to obtain a subject-level FBN representation  $\mathbf{h}_o \in \mathbb{R}^{1 \times d}$ :

$$\mathbf{h}_o = \sum_{v=1}^M \tilde{\alpha}_m \mathbf{h}_m \quad (5)$$

Through this mechanism, the model adaptively emphasizes disease-relevant ROIs while suppressing less informative nodes during feature aggregation. As a result, a more robust and discriminative individual FBN representation  $\mathbf{h}_o$  is obtained, which is subsequently used as the subject-level feature for population graph.

## 2.3. Disentangled dual-channel population graph learning

In multi-site neuroimaging studies, individual representations often contain both disease-related discriminative patterns and non-disease

variations introduced by scanning parameters and site differences. These factors are often highly entangled, limiting the model's ability to learn reliable pathological representations (Lin et al., 2023). Disentangled representation learning separates different sources of variation into distinct latent subspaces, allowing disease-related and site-related information to be modeled independently (Liu et al., 2022). Therefore, we propose a DDPG learning framework. Through parallel disease and site channels, DDPG separates disease-related information from site-related variations in the population topology. This explicitly models and suppresses site-related noise, enabling the disease channel to capture more biologically meaningful pathological representations.

### 2.3.1. Feature disentanglement and space projection

The model first disentangles  $\mathbf{h}_o$  to capture latent disease-related and site-related information. Specifically, two parameter-independent projection heads, denoted as  $P_d$  and  $P_s$ , are introduced to map the representation into two functionally separated latent subspaces. For the  $i$ th subject, the feature  $\mathbf{h}_o^i$  is projected by  $P_d$  into the disease-related space to obtain the disease feature  $\mathbf{h}_d^i \in \mathbb{R}^{d_e}$ :

$$\mathbf{h}_d^i = \sigma\left(\text{BN}\left(\mathbf{W}_d(\mathbf{h}_o^i)^\top + \mathbf{b}_d\right)\right) \quad (6)$$

where  $\mathbf{W}_d \in \mathbb{R}^{d_o \times d_e}$  and  $\mathbf{b}_d \in \mathbb{R}^{d_e}$  are learnable parameters, and BN denotes batch normalization.  $d_e$  is the dimensionality of the disentangled latent space.

Similarly, the site-specific feature  $\mathbf{h}_{site}^i \in \mathbb{R}^{d_e}$  is obtained by projecting  $\mathbf{h}_o^i$  into the site-specific subspace using  $P_s$ :

$$\mathbf{h}_{site}^i = \sigma\left(\text{BN}\left(\mathbf{W}_s(\mathbf{h}_o^i)^\top + \mathbf{b}_s\right)\right) \quad (7)$$

where  $\mathbf{W}_s \in \mathbb{R}^{d_o \times d_e}$  and  $\mathbf{b}_s \in \mathbb{R}^{d_e}$  are learnable parameters.

This mechanism provides a structural basis for separating disease- and site-related components. However, architectural separation alone does not ensure statistical independence. Since both projection heads share the same input  $\mathbf{h}_o^i$  and are jointly optimized, their latent subspaces may remain statistically dependent. Therefore, to further strengthen latent disentanglement, we introduce the Hilbert–Schmidt Independence Criterion (HSIC) as a regularization constraint. HSIC is a kernel-based statistical measure used to quantify the statistical dependence between two random variables (Gretton et al., 2005). Therefore, we minimize the HSIC value between the representation  $\mathbf{h}_d$  and  $\mathbf{h}_{site}$  to reduce their statistical dependence (Wang et al., 2024a):

$$\mathcal{L}_{\text{ortho}} = \text{HSIC}(\mathbf{H}_d, \mathbf{H}_s) = \frac{1}{(b-1)^2} \text{Tr}(\mathbf{K}_d \mathbf{H}_c \mathbf{K}_s \mathbf{H}_c) \quad (8)$$

where  $b$  denotes the number of samples in the current batch.  $\mathbf{H}_d = [\mathbf{h}_d^1, \dots, \mathbf{h}_d^b]^\top$  and  $\mathbf{H}_s = [\mathbf{h}_{site}^1, \dots, \mathbf{h}_{site}^b]^\top$  denote the disease-related and site-related representations, respectively.  $\mathbf{K}_d, \mathbf{K}_s \in \mathbb{R}^{b \times b}$  represent the kernel matrices computed from  $\mathbf{H}_d$  and  $\mathbf{H}_s$ , respectively, using the Gaussian radial basis function kernel.  $\mathbf{H}_c = \mathbf{I}_b - \frac{1}{b} \mathbf{1}\mathbf{1}^\top$  denotes the centring matrix, where  $\mathbf{I}_b$  is the identity matrix and  $\mathbf{1}\mathbf{1}^\top$  is an all-one matrix used to remove the mean offset in the kernel feature space.

### 2.3.2. Dual-channel population graph construction

Separating individual representations only in the feature space is insufficient, since site effects in multi-center data often manifest as structural differences in population-level distributions. Therefore, we further model disease- and site-related relational structures at the population level. Specifically, each subject is represented as a node, with features derived from the disentangled representations. Accordingly, we construct a disease graph  $G_{dis}$  to capture disease-discriminative associations and a site graph  $G_{site}$  to characterize site-related structural patterns.

**Disease Channel Graph Construction.** The disease graph  $G_{dis} = (V_p, A_{dis})$  integrates disease-related representations with non-imaging information to model inter-subject associations. Specifically, for each node  $i$ , its

disease-related representation  $\mathbf{h}_d^i$  is concatenated with the phenotypic feature vector  $\mathbf{p}_i$  to form the input for edge-weight estimation:

$$\mathbf{z}_d^i = [\mathbf{h}_d^i \parallel \mathbf{p}_i] \quad (9)$$

where  $\mathbf{p}_i = [\text{age}_i, \text{sex}_i]$  to describe the subject's non-imaging attributes, with age normalized as a continuous feature and sex encoded as an integer variable.

Based on the fused multimodal representations, the edge weight  $a_{dis}^{ij}$  between nodes  $i$  and  $j$  in the disease channel is adaptively estimated through the Disentangled Phenotypic Similarity Generator (DPSG). Specifically, a learnable nonlinear parser  $\phi(\cdot)$  projects the features into a latent space, and a temperature-scaled cosine similarity is then used to compute the connection strength:

$$a_{dis}^{ij} = \text{Sigmoid}\left(\tau_{dis} \cdot \frac{\phi(\mathbf{z}_d^i) \cdot \phi(\mathbf{z}_d^j)}{|\phi(\mathbf{z}_d^i)| |\phi(\mathbf{z}_d^j)|}\right) \quad (10)$$

where  $\phi(\cdot)$  is an MLP network, and  $\tau$  denotes the learnable temperature scale parameter.

In this way, the disease channel captures shared pathological patterns across subjects under the guidance of non-imaging phenotypic information.

**Site Channel Graph Construction.** In parallel, the site channel models systematic variations across scanning sites. Specifically, the site-related representation  $\mathbf{h}_{site}$  is constructed with the site label  $s_i$  to form  $\mathbf{z}_s^i = [\mathbf{h}_{site}^i \parallel s_i]$ , where  $s_i$  denotes the site index of subject  $i$ . DPSG then estimates the site-channel edge weights to construct the site graph  $G_{site} = (V_p, A_{site})$ :

$$a_{site}^{ij} = \text{Sigmoid}\left(\tau_s \cdot \frac{\phi_s(\mathbf{z}_s^i) \cdot \phi_s(\mathbf{z}_s^j)}{|\phi_s(\mathbf{z}_s^i)| |\phi_s(\mathbf{z}_s^j)|}\right) \quad (11)$$

where  $\phi_s(\cdot)$  is an MLP network.  $\tau_s$  represents a learnable temperature scale parameter.

Through this strategy, disease- and site-related relationships are explicitly disentangled at the structural level. Consequently, population information propagates within independent semantic spaces, providing a clear and stable structural foundation for subsequent learning.

### 2.3.3. Dual-channel ChebNet learning

After constructing DDPG, node information is propagated over both topologies to capture population-level relationships. To this end, we employ the Chebyshev Graph Convolutional Network (ChebNet) (Jiang et al., 2019) for graph learning in both channels. Compared with conventional spatial-domain GCNs, ChebNet approximates spectral graph filters using  $K$ -order Chebyshev polynomials, enabling the model to capture non-local population interactions across  $K$ -hop neighborhoods. This supports long-range dependency modeling and alleviates over-smoothing in deep graph networks. In DDPG, the disease and site channels learn their corresponding representations in parallel while maintaining disentangled feature spaces. For each channel, the normalized graph Laplacian is defined as:

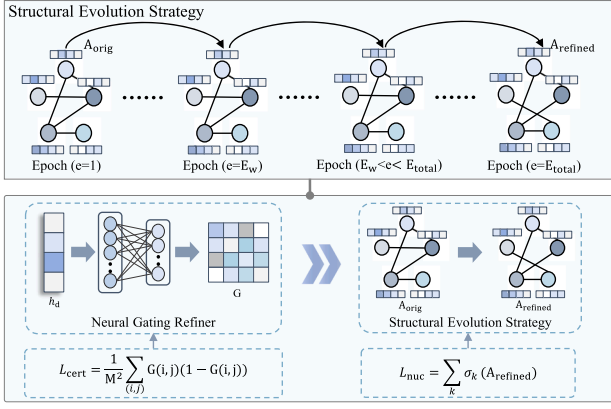
$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2} \quad (12)$$

where  $\mathbf{A} \in \{\mathbf{A}_{dis}, \mathbf{A}_{site}\}$  is the channel-specific adjacency matrix,  $N$  represents the total number of subjects,  $\mathbf{D}$  is its degree matrix, and  $\mathbf{I} \in \mathbb{R}^{N \times N}$  is the identity matrix.

Accordingly, we obtain the Laplacian matrices  $\mathbf{L}_{site}$  and  $\mathbf{L}_{dis}$  for the two channels. The propagation rule of the  $l$ th ChebNet layer is formulated as:

$$\mathbf{Z}^{(l+1)} = \sigma\left(\sum_{k=0}^{K-1} T_k(\mathbf{L}) \mathbf{Z}^{(l)} \Theta_k\right) \quad (13)$$

where  $\Theta_k \in \mathbb{R}^{d_l \times d_{l+1}}$  is learnable parameters.  $T_k(\cdot)$  is the  $k$ th Chebyshev polynomial. In the dual-channel setting,  $\mathbf{Z}^{(l)} \in \mathbb{R}^{N \times d_l}$  denotes the node



**Fig. 2.** Overview of the PDRM module. The disease-discriminative representation learned from the population graph is projected back into the edge space of individual FBNs through a neural gating refiner, and dynamically fused with the original adjacency matrix to obtain a refined brain graph.

feature matrix at layer  $l$ , corresponding to  $\mathbf{H}_d^{(l)}$  in the disease channel and  $\mathbf{H}_{site}^{(l)}$  in the site channel, respectively.

After stacking  $L$  layers, the enhanced representations  $\mathbf{Z}_d$  and  $\mathbf{Z}_s$  are obtained for downstream discrimination and joint optimization.

#### 2.3.4. Multi-task optimization

To improve disease identification and feature disentanglement, we formulate a multi-task joint optimization objective. It integrates disease and site identification, guiding each channel to learn task-specific features while reducing site-related interference in disease representations.

In the disease channel, disease identification encourages the learning of pathology-related representations. For binary identification, the disease loss  $\mathcal{L}_{dis}$  is defined as:

$$\mathcal{L}_{dis} = -\frac{1}{n} \sum_{i=1}^n \sum_{c=1}^2 y_{i,c} \log \hat{y}_{i,c} \quad (14)$$

where  $y_{i,c}$  denotes the ground-truth label of the  $i$ th sample for class  $c$ , and  $\hat{y}_{i,c}$  represents the predicted probability.

Meanwhile, site identification is introduced to guide the site graph to capture site-related variations. The site identification loss is defined as:

$$\mathcal{L}_{site} = -\frac{1}{n} \sum_{i=1}^n \sum_{s=1}^S y_{i,s}^{site} \log \hat{y}_{i,s}^{site} \quad (15)$$

where  $y_{i,s}^{site}$  denotes the ground-truth site label of the  $i$ th sample, and  $\hat{y}_{i,s}^{site}$  represents the predicted site probability.  $S$  is the number of scanning sites.

Finally, the overall optimization objective is defined as:

$$\mathcal{L}_{pop} = \mathcal{L}_{dis} + \lambda_1 \mathcal{L}_{site} + \lambda_2 \mathcal{L}_{ortho} \quad (16)$$

where  $\lambda_1$  and  $\lambda_2$  are trade-off hyperparameters balancing the contributions of different objectives.

#### 2.4. Population-guided dynamic refinement mechanism for individual brain networks

Although the initial FC matrix can capture the spatial topology between brain regions, individual FBNs may still contain pathology-irrelevant spurious connections caused by physiological interference and scanner noise (Zhang et al., 2026). To address this, we propose the PDRM which leverages stable disease-discriminative population representations learned at DDPG to refine the topological structure of individual brain graphs in a reverse correction manner, as shown in Fig. 2.

##### 2.4.1. Neural gating refiner

After disentanglement and population-level representation learning, the disease representation  $\mathbf{h}_d$  captures stable pathological patterns shared across subjects. To further enhance disease-discriminative topology modeling, we design a gating generator that maps  $\mathbf{h}_d$  back to the edge space of individual brain graphs. Specifically, the representation  $\mathbf{h}_d$  is fed into an MLP to generate a gating vector  $\mathbf{g} \in \mathbb{R}^{M^2}$ :

$$\mathbf{g} = \text{Sigmoid}(\text{MLP}(\mathbf{h}_d)) \quad (17)$$

The vector  $\mathbf{g}$  is then reshaped into a gating matrix  $\mathbf{G} \in \mathbb{R}^{M \times M}$ . The gating matrix adaptively regulates individual FCs to preserve task-relevant connection patterns and reduce the influence of task-irrelevant noise connections, thereby refining individual FBN topology.

##### 2.4.2. Structural evolution strategy

Although matrix  $\mathbf{G}$  provides task-driven topology, directly replacing the original FC structure may cause unstable training and biologically meaningful prior loss. Inspired by dynamic topology learning (Zhang et al., 2025), we dynamically integrate the original FC topology with the matrix  $\mathbf{G}$  to construct a refined individual adjacency matrix, thereby retaining the FC-based structural priors while incorporating task-guided adjustments for individual topology refinement:

$$\mathbf{A}_{refined} = \alpha(e) \cdot \mathbf{A}_{orig} + (1 - \alpha(e)) \cdot \mathbf{G} \quad (18)$$

The fusion coefficient  $\alpha(e)$  dynamically balances the original FC topology and the learned gating topology during training. It gradually shifts the model from relying on the original FC prior to incorporating task-guided topology adjustments, which is defined as:

$$\alpha(e) = \begin{cases} 1.0, & e \leq E_w \\ \max\left(0.1, 1.0 - k \cdot \frac{e - E_w}{E_{total}}\right), & e > E_w \end{cases} \quad (19)$$

where  $e$  denotes the current training epoch,  $E_w$  denotes the warm-up period,  $E_{total}$  denotes the total number of training epochs, and  $k$  is the decay control coefficient.

In the early stage,  $\alpha(e) = 1.0$  preserves the original FC topology for stable representation learning. As training progresses,  $\alpha(e)$  gradually decays, increasing the contribution of the learned gating topology. This process refines individual brain graphs by integrating task-guided adjustments while reducing the influence of task-irrelevant noise.

##### 2.4.3. Structural constraint

To preserve the modular organization of FBNs while supporting task-relevant topology refinement, we introduce two structural constraints. First, certainty regularization term encourages decisive edge modulation in  $\mathbf{G}$ :

$$\mathcal{L}_{cert} = \frac{1}{M^2} \sum_{i,j} \mathbf{G}(i,j)(1 - \mathbf{G}(i,j)) \quad (20)$$

Since each  $G_{ij} \in [0, 1]$  can be interpreted as the parameter of a Bernoulli edge-selection variable,  $G_{ij}(1 - G_{ij})$  corresponds to its variance and thus serves as a surrogate for uncertainty. Minimizing this term pushes the gates toward binary states, thereby improving the stability of topology refinement.

Furthermore, FBNs typically exhibit community and modular structures (Varoquaux et al., 2010). This implies that the adjacency matrix should exhibit low-rank properties in a mathematical sense. Therefore, we impose a nuclear norm constraint on the matrix  $\mathbf{A}_{refined}$  to enhance the biological plausibility of structural learning (Qiao et al., 2016). The nuclear norm regularization term is defined as:

$$\mathcal{L}_{nuc} = \|\mathbf{A}_{refined}\|_* = \sum_k \sigma_k(\mathbf{A}_{refined}) \quad (21)$$

where  $\sigma_k(\mathbf{A}_{refined})$  denotes the singular values of the matrix.

**Table 1**  
Demographic information of the dataset.

| Variable           | HC           | MDD          |
|--------------------|--------------|--------------|
| Number             | 765          | 821          |
| Male/Female        | 312/453      | 299/522      |
| Average age(years) | 33.26 ± 9.93 | 33.61 ± 9.61 |

Jointly minimizing the  $\mathcal{L}_{cert}$  and  $\mathcal{L}_{nuc}$  promotes stable topology refinement and reduces uncertain edge adjustments. Finally, the overall structural regularization term is defined as:

$$\mathcal{L}_{struct} = \lambda_3 \mathcal{L}_{cert} + \lambda_4 \mathcal{L}_{nuc} \quad (22)$$

where  $\lambda_3$  and  $\lambda_4$  denote weighting coefficient.

#### 2.4.4. Total loss

To balance different learning objectives, the above components are integrated into a unified multi-task optimization framework, with the final total loss defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{pop} + \mathcal{L}_{struct} \quad (23)$$

### 3. Experiments

#### 3.1. Dataset and preprocessing

We conducted a binary identification experiment to distinguish patients with MDD from healthy controls (HCs) on the publicly available REST-meta-MDD dataset. Following the official preprocessing pipeline (Yan et al., 2019), the raw data were standardized, and samples with missing key information or poor image quality were excluded. Ultimately, 1586 subjects were included in the analysis. The demographic characteristics of all participants are summarized in Table 1. The rs-fMRI data of each subject in the REST-meta-MDD dataset had already undergone standard preprocessing using the DPARSF (Yan & Zang, 2010). In this study, we directly used the data preprocessed through the official pipeline for subsequent analyses. According to the preprocessing scheme proposed by Yan et al. (2019), the main steps included removal of initial time points, slice-timing correction, head motion correction, spatial registration, normalization, nuisance covariate regression, and band-pass filtering at 0.01-0.10 Hz. In addition, we extracted the mean time series of brain regions based on the Dosenbach et al. (2010).

#### 3.2. Experimental setup

To evaluate the effectiveness of the proposed method on the REST-meta-MDD dataset, a random subject-level five-fold cross-validation was adopted to improve the robustness and reliability of the experimental results. Accuracy (ACC), sensitivity (SEN), specificity (SPE), and F1-score (F1) were selected as evaluation metrics. All experiments were implemented in the PyTorch framework and conducted on an NVIDIA GeForce RTX 5060 Ti GPU.

To validate the hyperparameter settings, we conducted tuning experiments on the training set. By comparing the identification performance, training stability, and relative contributions of different loss terms under different parameter combinations, we finally determined the current hyperparameter values. For individual graph construction, the FC threshold was set to  $\tau = 0.3$ . This threshold was used to remove weak and potentially unstable FCs while preserving relatively reliable inter-regional connections, thereby reducing noisy-edge effects on subsequent graph learning. Subsequently, individual brain graphs were encoded using a one-layer GIN with a dropout rate of 0.2. For dual-channel learning, a three-layer ChebNet was adopted, with the Chebyshev polynomial order set to  $K = 4$ , to capture higher-order neighborhood relationships in the population graph. In PDRM,  $\alpha(e)$  was scheduled using a staged linear decay strategy to promote model convergence and improve training

stability. This strategy was selected based on a comparison of multiple  $\alpha(e)$  scheduling schemes during five-fold cross-validation, where the variance of validation loss and the standard deviation of classification performance across repeated experiments with different random initializations were used as quantitative stability measures. Specifically, the first 20 epochs were used as a warm-up stage with  $\alpha(e) = 1$ , allowing the model to rely on the original FC structure for stable training. From epochs 20 to 50,  $\alpha(e)$  linearly decayed from 1.0 to 0.2. After epoch 50,  $\alpha(e)$  was fixed at 0.2. This strategy preserves the original brain network topology in the early training stage and gradually increases the contribution of the learned disease-discriminative structure to individual graph refinement, thereby avoiding training instability caused by premature reliance on the learned gating structure.

For the loss function weights, we determined the final values through preliminary experiments on the training set by considering both the magnitude of different loss terms and their influence on model performance. The disease identification loss  $\mathcal{L}_{diss}$  was used as the main task loss with a fixed weight of 1. The weight of the  $\mathcal{L}_{site}$  was set to  $\lambda_1 = 0.05$ , which helps model site-related information while avoiding excessive interference with disease identification. The weight of the  $\mathcal{L}_{ortho}$  was set to  $\lambda_2 = 0.5$  to promote statistical independence between disease- and site-related features. The weight of the  $\mathcal{L}_{cert}$  was set to  $\lambda_3 = 0.1$ , encouraging the gating matrix to make clearer edge selections and improving the stability and interpretability of structure refinement. The weight of the  $\mathcal{L}_{nuc}$  was set to  $\lambda_4 = 0.001$ , preserving the modular and low-rank properties of the refined brain graph while preventing this structural regularization term from dominating the identification objective. All weights were fixed across experiments. The model was trained with Adam for up to 100 epochs using an initial learning rate of 0.0001. Early stopping was used to prevent overfitting, terminating training after 30 consecutive non-improving validation epochs. In addition, for population graph construction, we used a fold-wise cross-validation strategy. In each fold, the training population graph was constructed solely from training subjects, with its nodes, candidate edges, and edge weights strictly restricted to the training set to eliminate train-test information leakage.

#### 3.3. Comparison with other works

To evaluate the proposed method, we compared DPGR-Net with traditional machine learning methods and representative GNN-based methods on the REST-meta-MDD dataset. All reproduced baselines and DPGR-Net were evaluated on the same ComBat-harmonized (Johnson et al., 2007) data for a fair comparison, with the best results highlighted in bold, as shown in Table 2. DPGR-Net achieved the best overall performance, demonstrating its effectiveness and robustness for multi-site MDD identification.

Specifically, the traditional machine learning method SVM showed the weakest performance, suggesting that shallow features and linear decision boundaries are insufficient to capture the complex brain network organization patterns. In contrast, GNN-based methods performed better, demonstrating the effectiveness of graph-structured modeling for MDD identification. Individual graph methods, such as BrainGNN and RGTNet, capture subject-specific brain topology but show considerable performance variation because they do not explicitly model cross-subject relationships. Among them, DSAM achieved the best performance among individual graph methods, which may be attributed to its ability to directly learn task-relevant FC from raw rs-fMRI time series while jointly characterizing the temporal dynamics and spatial dependencies of brain networks, thereby yielding more discriminative individual brain graph representations. It is also worth noting that although DG-DMSGCN introduces site information into individual brain graph modeling, its performance remains lower than that of some individual brain graph models that do not use non-imaging information. This suggests that simply incorporating site information, without an effective mechanism to model or correct inter-site distribution shifts, may be insufficient to alleviate multi-site heterogeneity. If site information is not

**Table 2**

Performance comparison with other methods. IndG denotes individual graph methods, PopG denotes population graph methods, and Joint denotes methods that jointly model individual and population graphs.

| Method                               | IndG | PopG | Joint | Non-imaging | ACC (%)             | SEN (%)             | SPE (%)             | F1 (%)              |
|--------------------------------------|------|------|-------|-------------|---------------------|---------------------|---------------------|---------------------|
| SVM (Cortes & Vapnik, 1995)          | ×    | ×    | ×     | ×           | 57.57 ± 3.12        | 58.70 ± 3.55        | 56.34 ± 6.01        | 58.88 ± 2.66        |
| GAT (Veličković et al., 2018)        | ✓    | ×    | ×     | ×           | 60.84 ± 2.96        | 62.24 ± 6.36        | 59.35 ± 5.72        | 62.07 ± 3.81        |
| GCN (Jiang et al., 2019)             | ✓    | ×    | ×     | ×           | 61.82 ± 2.60        | 60.79 ± 5.74        | 54.64 ± 3.85        | 59.76 ± 3.52        |
| PopulationGCN (Parisot et al., 2017) | ×    | ✓    | ×     | ✓           | 63.50 ± 2.62        | 61.80 ± 7.32        | 60.85 ± 3.96        | 62.14 ± 4.00        |
| Hi-GCN (Jiang et al., 2020)          | ×    | ×    | ✓     | ✓           | 62.26 ± 3.24        | 64.24 ± 6.52        | 63.79 ± 5.63        | 63.86 ± 4.15        |
| EV-GCN (Huang & Chung, 2020)         | ×    | ✓    | ×     | ✓           | 76.23 ± 8.20        | 82.33 ± 4.85        | 70.46 ± 2.76        | 78.21 ± 5.21        |
| BrainGNN (Li et al., 2021)           | ✓    | ×    | ×     | ×           | 62.45 ± 3.11        | 65.23 ± 2.25        | 60.33 ± 3.98        | 61.39 ± 7.07        |
| DPGC (Lin et al., 2023)              | ×    | ✓    | ×     | ✓           | 90.61 ± 1.62        | 90.87 ± 3.99        | 87.33 ± 3.37        | 90.90 ± 1.55        |
| DG-DMSGCN (Cui et al., 2023)         | ✓    | ×    | ×     | ✓           | 63.38 ± 2.31        | 77.58 ± 2.97        | 66.43 ± 1.64        | 70.85 ± 2.89        |
| RGNet (Wang et al., 2024b)           | ✓    | ×    | ×     | ×           | 70.09 ± 3.50        | 69.33 ± 3.58        | 67.18 ± 2.27        | 69.34 ± 2.23        |
| BrainDAS (Song et al., 2024)         | ✓    | ×    | ×     | ✓           | 86.67 ± 5.80        | 86.86 ± 7.04        | 85.74 ± 9.32        | 86.36 ± 6.21        |
| DSAM (Thapaliya et al., 2025b)       | ✓    | ×    | ×     | ×           | 77.01 ± 1.54        | 75.28 ± 2.38        | 76.09 ± 4.93        | 76.48 ± 3.47        |
| FC-HGNN (Gu et al., 2025)            | ×    | ×    | ✓     | ✓           | 89.27 ± 2.13        | 90.85 ± 2.46        | <b>87.58 ± 2.98</b> | 89.77 ± 1.95        |
| <b>DPGR-Net(ours)</b>                | ×    | ×    | ✓     | ✓           | <b>92.37 ± 1.93</b> | <b>97.20 ± 2.02</b> | 87.19 ± 3.68        | <b>92.96 ± 1.71</b> |

properly utilized, it may even introduce additional noise and weaken model performance. In contrast, BrainDAS further exploits site labels from the perspective of graph domain adaptation. By encoding source-site information as domain indicators to guide dynamic kernel generation and performing graph-level domain adaptation, BrainDAS can better align individual brain graphs across sites, leading to improved identification performance.

Population graph methods, such as PopulationGCN and EV-GCN, generally outperform early individual-graph models by exploiting cross-subject relationships. In particular, EV-GCN improves identification through adaptive population graph construction, highlighting the importance of cross-subject relationship modeling in multi-site MDD identification. DPGC further addresses multi-site heterogeneity by disentangling site-invariant and site-specific representations. Its competitive performance demonstrates the effectiveness of disentanglement for reducing site-related heterogeneity and improving MDD identification. However, although population graph methods achieve strong identification performance by modeling cross-subject relationships, their interpretability is relatively limited because they mainly focus on subject-level associations rather than explicit brain network structures. In contrast, joint individual-population graph methods can achieve competitive performance while preserving more interpretable brain-region-level information. For example, FC-HGNN integrates intra- and inter-hemispheric information interactions and models cross-subject relationships with heterogeneous population graphs, thereby combining individual brain network features with population-level relational information. In comparison, DPGR-Net further disentangles disease-related information from site-related variation at the population level, and suppresses the interference of site-related noise on individual topological structure learning through PDRM for individual brain graphs, thereby achieving superior multi-site MDD identification. It is worth noting that DPGR-Net achieves higher SEN than SPE, indicating greater sensitivity to MDD but a higher risk of misclassifying HCs. This may be related to the design objective of DPGR-Net, since the disentanglement strategy and PDRM allow the model to emphasize MDD-discriminative features, thereby capturing abnormal brain functional patterns with greater sensitivity. Although DPGR-Net has slightly lower SPE than FC-HGNN, its higher F1 indicates better overall performance while maintaining comparable HC identification.

### 3.4. Ablation studies

#### 3.4.1. Ablation study on modules

In this ablation study, we systematically analyzed the impact of different key components on the overall performance of the model by gradually removing them. The experimental results are shown in the Table 3, and each component of the model has a contribution to the performance.

**Table 3**

Ablation study results of different modules.

| Method                                           | ACC (%) | SEN (%) | SPE (%) | F1 (%) |
|--------------------------------------------------|---------|---------|---------|--------|
| w/o PDRM                                         | 82.21   | 75.86   | 89.02   | 73.08  |
| w/o $\mathcal{L}_{cert}$                         | 90.19   | 95.51   | 86.14   | 90.64  |
| w/o $\mathcal{L}_{nuc}$                          | 90.17   | 95.62   | 86.41   | 90.81  |
| w/o $\mathcal{L}_{cert}$ and $\mathcal{L}_{nuc}$ | 89.33   | 94.27   | 85.64   | 89.17  |
| w/o Disentanglement                              | 90.10   | 96.47   | 83.27   | 90.98  |
| w/o Site Channel                                 | 90.41   | 95.62   | 84.84   | 91.18  |
| DPGR-Net                                         | 92.37   | 97.20   | 87.19   | 92.96  |

Specifically, removing PDRM prevents the model from using stable population-level pathological information to refine individual brain topology, resulting in a clear performance drop. This indicates that the original FC matrices contain subject-specific noise and unstable connections, whereas PDRM suppresses such interference and strengthens disease-discriminative connections, thereby improving the reliability and discriminability of individual representations. We further evaluated the structural constraints within PDRM. The results show that removing either constraint leads to performance degradation, indicating that the improvement of PDRM depends on effective regularization during structural optimization. Removing  $\mathcal{L}_{cert}$  reduces ACC and F1 to 90.19% and 90.64%, respectively. This suggests that without the gate certainty constraint, some edge weights may remain ambiguous, weakening the key FC selection. Removing  $\mathcal{L}_{nuc}$  results in ACC and F1 of 90.17% and 90.81%, respectively, indicating that the lack of the low-rank constraint may lead to more scattered or redundant connections in the refined brain graph, thereby affecting the rationality of the overall brain network structure.

In addition, removing the site channel or the disentanglement module also leads to performance degradation to different extents. Specifically, the former means that only the disease channel is retained for population relationship modeling, while the latter removes the dual-projection mechanism and the HSIC independence constraint, causing the two channels to directly share the same readout features. This indicates that explicitly modeling site-related information is necessary in multi-site settings. The reason is that the original rs-fMRI features usually contain both disease-related pathological patterns and site effects. If they are directly used for identification without distinction, the model may fail to focus on disease-related patterns. Through the dual-channel population graph learning framework, the model can separately model disease-related information and site-related information. With the disentanglement strategy, the interference of site effects on disease representation learning can be further reduced, so that the learned features are more focused on task-relevant pathological patterns.

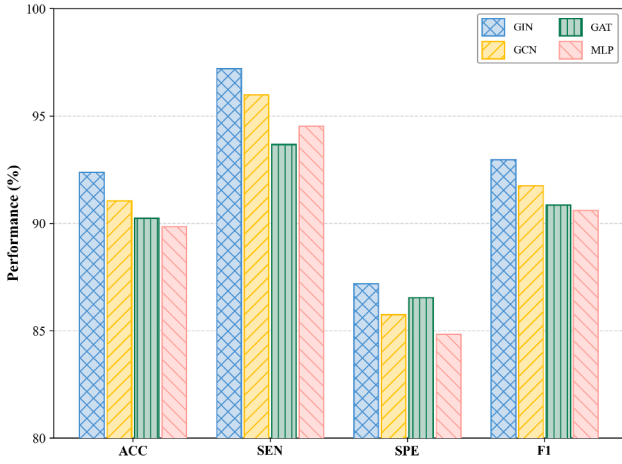


Fig. 3. Performance comparison of different encoders for individual brain graph representation learning.

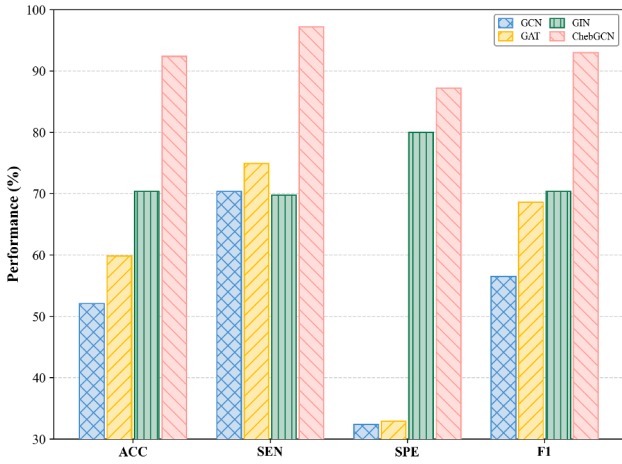


Fig. 4. Performance comparison of different graph convolutional layers in DDPG learning.

### 3.4.2. Ablation study on graph learning layers

Since the PDRM module demonstrated the most significant performance contribution in the ablation study, we further evaluated the effect of different backbone encoders by replacing GIN with GCN, Graph Attention Network (GAT), and MLP. The results are shown in the Fig. 3. Compared with GCN's simple neighborhood averaging, GIN improves the model's expressive power by introducing learnable parameters and MLP. In brain network analysis, this mechanism allows the model to capture finer node topology differences, thus generating more discriminative individual brain network representations and achieving optimal identification accuracy. MLP operates on node features independently without explicitly modeling the topological relationships among brain regions, making it difficult to capture the complex interaction patterns embedded in brain networks, which consequently leads to inferior performance. Notably, although GAT introduces an attention mechanism to enable adaptive neighborhood aggregation, the learned attention weights are susceptible to noisy connections, thereby compromising the stability of feature aggregation and limiting the model potential performance.

Second, to evaluate the influence of different graph convolutional layers in the population graph learning module, we replaced ChebNet with GCN, GAT, and GIN while keeping the other modules unchanged. As shown in Fig. 4, ChebNet achieves the best identification performance, whereas GCN and GAT exhibit lower overall performance, especially in terms of SPE. Although GIN outperforms GCN and GAT, it still performs worse than ChebNet. These results suggest that multi-center

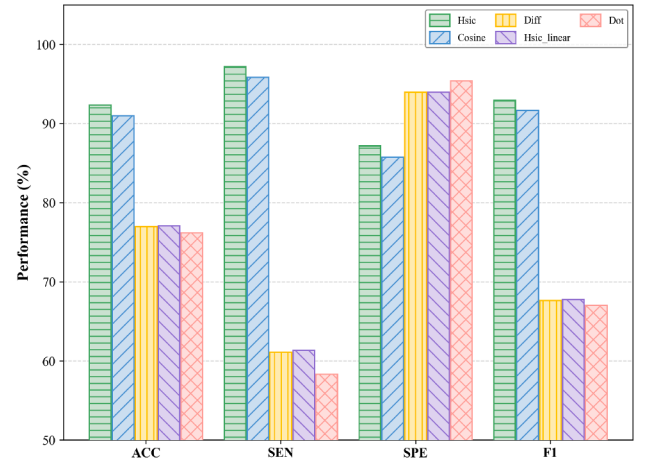


Fig. 5. Comparison of different disentanglement constraints.

population graphs involve high-order cross-subject dependencies that cannot be adequately captured by local first-order aggregation. GCN and GIN mainly aggregate information from first-order neighborhoods, while increasing their depth may lead to over-smoothing or unstable training. GAT adaptively weights neighboring nodes, but its information propagation remains largely confined to local neighborhoods. In contrast, ChebNet employs Chebyshev polynomials to aggregate multi-hop information within a single layer, enabling better modeling of high-order population relationships.

### 3.5. Evaluation of disentanglement regularization strategies

To evaluate feature disentanglement, we compared different independence constraints. The experimental results are shown in Fig. 5. The results demonstrate that the choice of disentanglement loss plays a critical role in determining the final identification performance. Among all compared strategies, the Gaussian-kernel HSIC achieved the best results, suggesting that the dependency between disease- and site-related representations is inherently nonlinear and cannot be adequately removed by simple linear constraints alone. In contrast, the cosine constraint yielded the second-best performance, indicating that constraining representation similarity only from the perspective of feature direction can partially alleviate the entanglement. However, since cosine similarity ignores feature magnitude and higher-order statistical dependencies, it can only encourage coarse-grained directional separation rather than eliminating deeper nonlinear dependencies. By comparison, DOT, differential covariance (Diff), and linear-kernel HSIC resulted in substantial performance degradation, characterized by high SPE but markedly lower SEN and F1. This phenomenon indicates that although these constraints enhance representation separation, they may also suppress discriminative disease-related information. This may result from their limited ability to model complex dependencies or from overly restrictive constraints that impair task-relevant feature learning.

### 3.6. Cross-site generalization evaluation

To evaluate the cross-site generalization ability of the proposed model, we conducted leave-one-site-out (LOSO) validation. In each round, one site was held out as the test set, while the remaining sites were used for training. This setting simulates a real-world clinical scenario in which a model trained on large-scale data from existing centers is applied to assist the diagnosis of patients from a new and previously unseen acquisition center.

The experimental results are presented in Table 4. Overall, DPGR-Net achieved good average performance across the 16 LOSO tests, demonstrating stable disease identification on most unseen sites and partial

**Table 4**  
identification performance and data distribution across different sites.

| Site | MDD:HC  | ACC(%) | SEN(%) | SPE(%) | F1(%) |
|------|---------|--------|--------|--------|-------|
| S1   | 73:73   | 92.47  | 100.00 | 84.93  | 92.99 |
| S2   | 16:14   | 70.00  | 81.25  | 57.14  | 74.29 |
| S7   | 35:37   | 87.50  | 97.14  | 78.38  | 88.31 |
| S8   | 49:60   | 85.32  | 85.71  | 85.00  | 84.00 |
| S9   | 48:48   | 92.71  | 95.83  | 89.58  | 92.93 |
| S10  | 45:26   | 88.73  | 86.67  | 92.31  | 90.70 |
| S11  | 20:17   | 94.59  | 100.00 | 88.24  | 95.24 |
| S13  | 20:16   | 80.56  | 95.00  | 62.50  | 84.44 |
| S14  | 61:32   | 88.17  | 88.52  | 87.50  | 90.76 |
| S15  | 30:37   | 74.63  | 100.00 | 54.05  | 77.92 |
| S17  | 41:41   | 92.68  | 97.56  | 87.80  | 93.02 |
| S19  | 18:31   | 77.55  | 94.44  | 67.74  | 75.56 |
| S20  | 246:225 | 95.97  | 97.15  | 94.67  | 96.18 |
| S21  | 79:65   | 94.44  | 100.00 | 87.69  | 95.18 |
| S22  | 18:20   | 81.58  | 94.44  | 70.00  | 82.93 |
| S23  | 22:23   | 88.67  | 95.45  | 78.26  | 87.50 |

robustness to multicenter distribution shifts. Specifically, Site 20 achieved the best identification performance, which may be partly attributed to its relatively large sample size and balanced distribution of patients with MDD and HCs. A larger test sample size is generally less sensitive to a small number of misclassified subjects, while a balanced class distribution contributes to more stable evaluation results. Similarly, other sites with larger sample sizes generally obtained higher ACC and F1, further indicating that sample size and data distribution stability play important roles in cross-site generalization evaluation. In contrast, Site 2, Site 15, and Site 19 showed relatively lower performance, which may be related to their smaller test sample sizes, imbalanced class distributions, and site-specific data characteristics. For test sites with fewer samples, a small number of misclassified subjects may lead to noticeable fluctuations in ACC, SEN, SPE, and F1, making the evaluation results less stable than those of larger sites. Moreover, differences in scanner types, acquisition protocols, and population characteristics may increase the distribution gap between the test site and the training sites, making it difficult for the model to adapt to unseen site-specific patterns. Meanwhile, an imbalanced distribution between patients with MDD and healthy controls within a site may also bias the predictions toward the majority class, thereby further affecting the model performance on that site.

While Table 4 focuses on the site-wise performance of DPGR-Net, Table 5 compares its average LOSO performance with that of representative baseline methods under identical LOSO validation settings. Overall, the proposed method achieved favorable results in terms of ACC, SEN, and F1, indicating that it can more effectively improve overall identification performance and disease-sample identification ability in cross-site testing scenarios. Specifically, the clear improvement in SEN suggests that the model can more effectively capture MDD-related discriminative features, thereby improving patient identification. Meanwhile, the improved F1 indicates a better trade-off between disease-sample identification and overall classification balance. To further assess the statistical reliability of the above performance differences, we conducted paired t-tests based on the site-wise results from the same 16 held-out sites. Compared with DG-DMSGCN and EV-GCN, DPGR-Net achieved statistically significant improvements across all metrics ( $p < 0.05$ ), indicating that it has a relatively stable overall cross-site advantage over these two baseline methods. Compared with the more competitive baseline model FC-HGNN, DPGR-Net also achieved statistically significant improvements in ACC and SEN, suggesting that its advantages are mainly reflected in overall classification accuracy and MDD sample identification. By contrast, the differences between the two methods in SPE and F1 were not statistically significant. This indicates that the average differences between the two methods in these metrics were relatively small, and that the cross-site performance differences did not exhibit a sufficiently

consistent trend. In terms of the average results, DPGR-Net achieved a slightly lower SPE but a slightly higher F1 than FC-HGNN, suggesting that the two methods performed comparably in healthy control identification and overall classification performance. Taken together, the cross-site performance gains of DPGR-Net were mainly reflected in ACC and SEN, demonstrating more stable advantages in overall classification accuracy and MDD sample identification. Therefore, future work will focus on improving model specificity while preserving MDD sample identification ability, as well as enhancing its adaptability to distributional variations at unseen sites and the stability of its cross-site generalization.

## 4. Discussions

### 4.1. The interpretability analysis

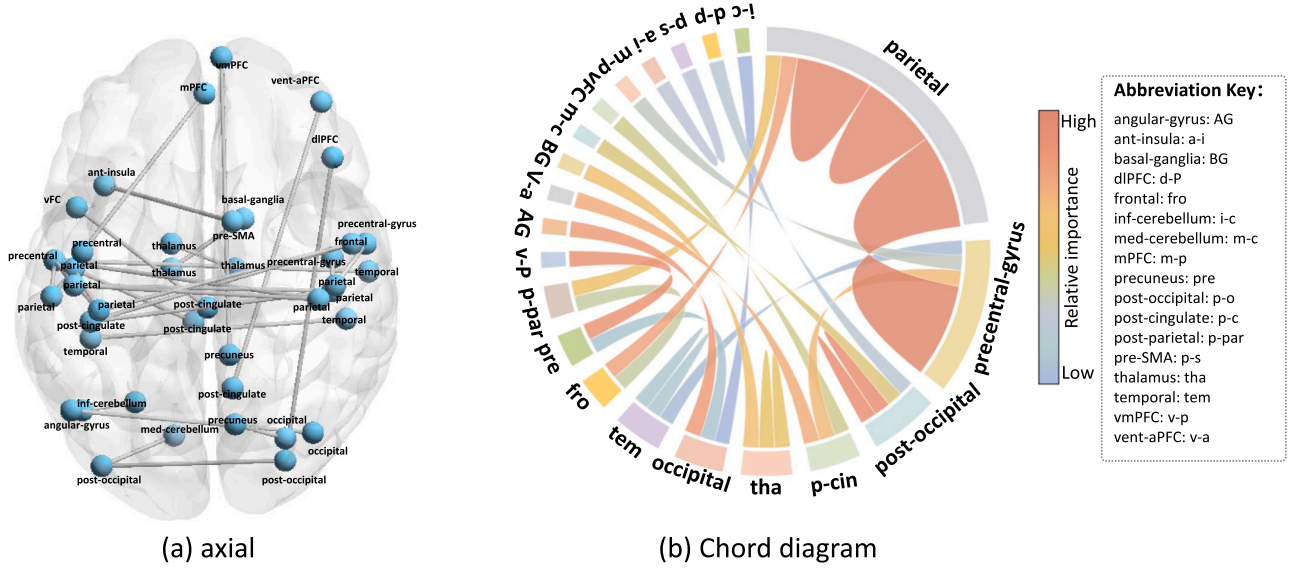
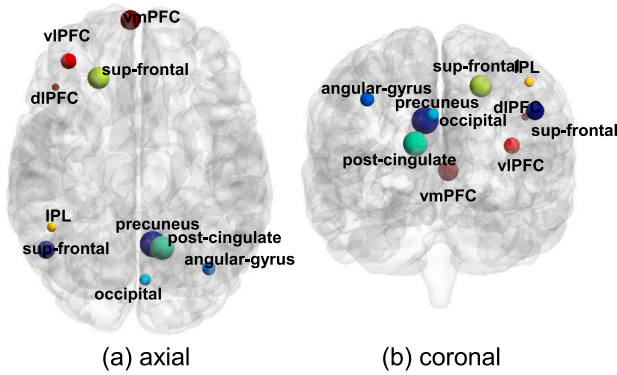
To explore the potential FC patterns captured by the model for MDD identification, we ranked the FCs according to the mean absolute group-wise differences in  $A_{refined}$  between the MDD and HC groups, and selected the top 30 connections with the largest differences for visualization analysis. As shown in the Fig. 6, their spatial distribution is displayed on the brain surface in Fig. 6 (a) and as their relative importance and inter-regional organization in the chord diagram in Fig. 6 (b). Taken together, these discriminative edges were not randomly distributed but were mainly concentrated in core pathological circuits implicated in depression. First, the model identified abnormal connections involving the mPFC, vmPFC, and precuneus, suggesting that it captured disruptions in DMN-related pathways. Previous studies have shown that abnormal functional connectivity in these regions is closely associated with altered self-referential processing, ruminative thinking, and negative emotional processing in MDD (Yan et al., 2019). Second, FCs involving the anterior insula, pre-SMA, and dlPFC exhibited high discriminative scores, indicating abnormal coordinated regulation between the salience network and the cognitive control network, which may underlie deficits in attentional allocation, emotion regulation, and cognitive control in MDD patients (Seeley et al., 2007). In addition, differences were observed in subcortical FC between the thalamus and basal ganglia, further supporting dysfunction within the cortico-basal ganglia-thalamo circuit. Previous evidence suggests that abnormalities in this circuit are closely related to disturbances in reward-related and motivational processing and psychomotor symptoms in depression (Hagen et al., 2025). Meanwhile, the repeated between-group differences involving the pre-central gyrus and parietal suggest that depression-related abnormalities may further extend to the sensorimotor network and its aberrant coupling with higher-order control systems (Javaheripour et al., 2021). Overall, the key connectivity patterns identified by the model are more likely to reflect coordinated abnormalities across the DMN, the salience and cognitive control systems, as well as subcortical motor-reward circuits.

To further assess the neurobiological relevance of DPGR-Net, we identified the top 10 most contributive brain regions based on the ROI-level attention weights learned by the SERO module. To further validate the discriminative contribution of the model-identified brain regions, we performed an ROI random-removal analysis. The results showed that removing the top 10 ROIs identified by the model led to a significant decrease in ACC, with a larger performance drop than randomly removing 10 ROIs ( $p < 0.05$ ). This indicates that the identified ROIs make meaningful contributions to MDD identification. As shown in the Fig. 7, the discriminative information captured by the model is not scattered across several isolated brain regions, but is mainly concentrated in regions associated with the default mode network (DMN) and key nodes related to the frontoparietal control system. Specifically, the model's strong focus on core DMN regions, such as the precuneus, posterior cingulate, and vmPFC, suggests that DPGR-Net captures pathological patterns associated with self-referential processing, introspective activity, and rumination (Yan et al., 2019). Meanwhile, the prominent contributions of regions such as the dlPFC and superior frontal gyrus indicate that

**Table 5**

Average LOSO performance across the 16 held-out sites and paired statistical comparisons with DPGR-Net. The best average performance for each metric is shown in bold.

| Method    | ACC (%)                            | SEN (%)                            | SPE (%)                                           | F1 (%)                             |
|-----------|------------------------------------|------------------------------------|---------------------------------------------------|------------------------------------|
| DG-DMSGCN | 69.80 $\pm$ 6.80 ( $p < .01$ )     | 71.00 $\pm$ 9.65 ( $p < .01$ )     | 68.30 $\pm$ 9.05 ( $p < .01$ )                    | 71.25 $\pm$ 7.15 ( $p < .01$ )     |
| EV-GCN    | 71.92 $\pm$ 12.99 ( $p < .01$ )    | 74.35 $\pm$ 24.57 ( $p < .01$ )    | 69.49 $\pm$ 27.52 ( $p < .01$ )                   | 77.84 $\pm$ 14.30 ( $p < .01$ )    |
| FC-HGNN   | 83.04 $\pm$ 9.22 ( $p = .049$ )    | 89.77 $\pm$ 7.36 ( $p = .026$ )    | <b>80.92 <math>\pm</math> 5.54</b> ( $p = .362$ ) | 81.98 $\pm$ 6.21 ( $p = .059$ )    |
| DPGR-Net  | <b>86.60 <math>\pm</math> 7.74</b> | <b>94.32 <math>\pm</math> 5.75</b> | 79.11 $\pm$ 12.90                                 | <b>87.62 <math>\pm</math> 7.10</b> |

**Fig. 6.** Top 30 discriminative FCs for MDD identification.**Fig. 7.** Top 10 contributive regions in MDD identification.

the model captures critical neural representations related to cognitive control and executive function (Zhu et al., 2022). In addition, the involvement of the vlPFC further suggests that, beyond classical executive control processes, the model also identifies abnormalities in prefrontal regulatory mechanisms associated with response inhibition and the regulation of negative thoughts. Furthermore, the involvement of the angular gyrus, IPS, and IPL implies that depression-related abnormalities may also involve disrupted attentional allocation, multimodal information integration, and impaired coordination within the frontoparietal control system (Scolari et al., 2015).

#### 4.2. Visual verification of feature disentanglement

To further investigate the effectiveness of the proposed framework in representation learning and feature disentanglement, we visualized the disease-related features using t-SNE (van der Maaten &

Hinton, 2008). By projecting the high-dimensional disentangled features into a 2D space, we can evaluate the model's discriminative power. When colored by diagnosis labels (HC vs. MDD), the samples exhibit distinct clustering patterns with clear boundaries, as shown in Fig. 8(a), which strongly demonstrates that the model successfully captures highly discriminative pathological representations from complex brain networks. However, when the same features are colored by site labels, the samples from different acquisition sites are highly intertwined and uniformly distributed, without any site-specific clusters, as shown in Fig. 8(b). This distribution pattern indicates that the proposed disentanglement mechanism effectively suppresses the influence of non-pathological site effects on the feature space. These results demonstrate that the model captures discriminative features highly correlated with MDD pathology, ensuring its reliability in clinical environments.

#### 4.3. Limitations and future work

Although the proposed method achieved favorable performance in multi-center MDD identification, cross-center generalization remains challenging due to differences in sample size, class distribution, scanning protocols, scanner types, and population characteristics across centers. By disentangling disease- and site-related representations, the proposed method alleviates the influence of center heterogeneity to some extent. However, performance fluctuations were still observed on some held-out sites with small sample sizes or imbalanced class distributions, suggesting that further improving the model's adaptability to unseen center distributions remains important. Future work may incorporate domain adaptation, domain generalization, or uncertainty modeling strategies to enhance cross-center robustness.

In addition, age, sex, and site information were incorporated to construct population relationships. Among them, site labels can, to some

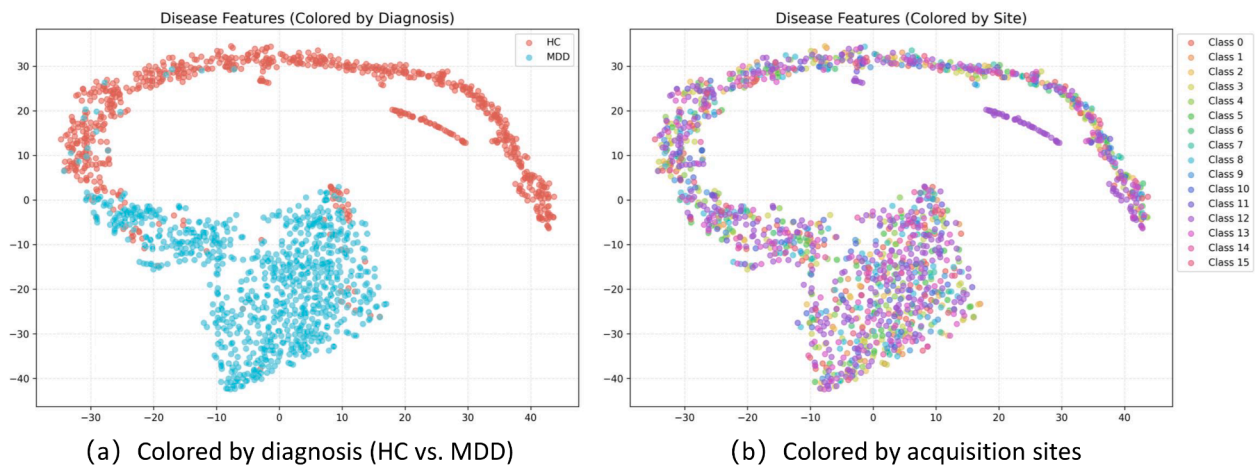


Fig. 8. T-SNE visualization of the learned disease-related features  $\mathbf{h}_d$  on the Rest-meta-MDD dataset.

extent, reflect the overall differences across centers in terms of scanner equipment, acquisition protocols, recruitment criteria, and sample composition. However, site labels provide only a coarse-grained representation of center heterogeneity, making it difficult to further distinguish the specific contributions of different factors to cross-center distribution shifts and performance fluctuations. Therefore, while the proposed method mitigates overall center effects to some extent, it has limited capacity for fine-grained modeling and interpretation of the sources of center heterogeneity. Meanwhile, due to the incompleteness of some clinical variables in the current dataset, this study was unable to systematically analyze the distributional differences across centers in clinical characteristics such as illness duration, disease stage, and medication status, or whether these differences affect brain functional connectivity patterns and model diagnostic performance. Therefore, future research will incorporate more comprehensive clinical information and imaging acquisition information to further analyze the effects of clinical heterogeneity and acquisition-related differences on model diagnostic performance, thereby improving the robustness and interpretability of the model in multi-center application scenarios.

## 5. Conclusion

In this study, we propose DPGR-Net, a disentangled population-guided graph neural network for MDD identification using multicenter rs-fMRI data. This study shows that separating disease-related information from site-related variations improves the robustness of multicenter MDD classification. Moreover, using population-level disease-discriminative representations to refine individual FC topology further enhances the model's discriminative ability. By constructing the DDPG module, DPGR-Net models disease-related associations and site-related differences in separate latent spaces, reducing the interference of site effects on disease representation learning. Furthermore, the proposed PDRM feeds stable population-level disease information back to individual brain graph refinement, enabling task-guided optimization of subject-specific FC topology. Experimental results on a public dataset demonstrate that the proposed method outperforms existing state-of-the-art approaches in multicenter MDD identification, validating its effectiveness and potential for practical application. In addition, DPGR-Net provides neurobiologically meaningful interpretability by identifying discriminative brain regions and aberrant functional connectivity patterns associated with MDD. Overall, this study provides a new perspective for the computer-aided diagnosis of mental disorders in multicenter neuroimaging settings and demonstrates potential value for improving clinical diagnostic accuracy and supporting medical decision-making.

## CRedit authorship contribution statement

**Xuan He:** Supervision, Methodology, Funding acquisition, Writing – review & editing; **Ting Mei:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis; **Huanhuan Du:** Data curation, Methodology, Validation; **Xinting Zhou:** Validation; **Jin-gran Xiang:** Visualization; **Zheng Liu:** Formal analysis.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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